

Sri Lanka Institute of Information Technology



Deep Learning-Based Speech Emotion Recognition for Supporting Emotional Well-Being of Adults Living Independently at Home

DECLARATION AND ACKNOWLEDGEMENT

DECLARATION

We hereby declare that this report, entitled “Deep Learning-Based Speech Emotion Recognition for Supporting Emotional Well-Being of Adults Living Independently at Home”, has been prepared as part of the requirements for SE4050 Deep Learning of the BSc (Hons) in Information Technology degree at the Sri Lanka Institute of Information Technology (SLIIT).

We declare that the work presented in this report is based on the development, implementation, experimentation, and analysis carried out by the project group. The experimental results, interpretations, figures, tables, and conclusions presented in this report are based on the experiments conducted using the selected dataset and the implemented deep learning models.

The CREMA-D dataset and relevant published literature have been appropriately acknowledged and cited. Open-source software libraries, frameworks, and other publicly available resources used during the development of the project have also been acknowledged where applicable. The project source code and development work are maintained in the associated GitHub repository.

Where artificial intelligence (AI)-assisted tools were used during the preparation of this report or the development process, such assistance was used as a supporting tool for activities such as language refinement, explanation, brainstorming, and technical guidance. The project group remains responsible for verifying the technical accuracy, originality, interpretation, and final content of this report.

We confirm that this report has been prepared in accordance with the academic and ethical requirements of the Sri Lanka Institute of Information Technology.

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We would also like to acknowledge the creators and contributors of the **CREMA-D (Crowd-sourced Emotional Multimodal Actors Dataset)** for making the dataset available for research purposes. The dataset provided the speech recordings required to investigate the proposed speech emotion recognition problem.

We acknowledge the open-source technologies and software libraries used in this project, including **Python, TensorFlow, Keras, NumPy, Pandas, Librosa, Scikit-learn, Matplotlib**, and other supporting tools used for data processing, model development, training, evaluation, and visualization.

The project source code, configurations, experimental implementations, and development history were maintained using **Git and GitHub**, which supported version control, collaboration, reproducibility, and traceability throughout the development process.

We also acknowledge the research papers, technical documentation, and other scholarly resources consulted during the design and development of the proposed system. These resources contributed to the selection of appropriate preprocessing techniques, feature representations, deep learning architectures, regularization methods, and evaluation strategies.

Where applicable, **AI-assisted tools** were used as supplementary resources during the development and documentation process. Such tools were used for activities including language improvement, clarification of technical concepts, brainstorming, and documentation assistance. All technically relevant information, experimental results, model configurations, and conclusions were reviewed and verified by the project group before being incorporated into the final report.

Finally, we would like to thank all individuals who provided technical, academic, and general support throughout the completion of this project.

ABSTRACT

Deep Learning-Based Speech Emotion Recognition for Supporting Emotional Well-Being of Adults Living Independently at Home

Speech contains acoustic information related to human emotional states, making Speech Emotion Recognition (SER) an important application of deep learning. However, accurate emotion recognition from speech remains challenging because of speaker variability, temporal characteristics, similarities between emotions, and difficulties in generalizing to unseen speakers. This study investigates supervised deep learning approaches for speaker-independent speech emotion recognition across six emotional categories: Anger, Disgust, Fear, Happy, Neutral, and Sad.

The study uses the CREMA-D (Crowd-sourced Emotional Multimodal Actors Dataset), containing 7,442 speech recordings from 91 actors. To reduce speaker-related data leakage and evaluate generalization to unseen speakers, the dataset was divided at actor level into 63 training, 13 validation, and 15 test actors. The audio recordings were preprocessed and converted into acoustic representations using MFCCs and Log-Mel spectrograms, with normalization statistics calculated from the training data only. SpecAugment was additionally applied during model training to improve robustness and reduce overfitting.

Six deep learning architectures were investigated and compared: BiLSTM with Attention, LSTM with Attention Pooling, CNN-BiGRU with Dual Attention, CNN-SE-BiLSTM with Multi-Head Attention, CNN-BiLSTM with Multi-Head Attention and SpecAugment, and TCN-MHA with Attention. The models were evaluated using accuracy, precision, recall, macro F1-score, and confusion matrices, with particular attention given to generalization and class-level errors. Among the evaluated experiments, the CNN-BiLSTM-Multi-Head Attention with SpecAugment model achieved the highest observed test performance, with 59.89% accuracy and 59.73% macro F1-score. The experiments also showed differences in generalization behavior across architectures, with some models exhibiting greater training-validation gaps and overfitting than others.

Overall, the results demonstrate that deep learning can effectively learn useful acoustic representations for speaker-independent speech emotion recognition, while also highlighting the challenges of emotion overlap, unseen-speaker generalization, and model complexity. The findings provide a foundation for future improvements through stronger regularization, larger and more diverse datasets, and multimodal approaches for emotion and well-being assessment.

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Problem Definition

Speech Emotion Recognition (SER) is formulated in this study as a supervised multiclass classification problem, where a speech recording is classified into one of six predefined emotion categories: **Anger, Disgust, Fear, Happy, Neutral, and Sad**. The main objective is to develop deep learning models that can learn meaningful acoustic patterns from speech and generalize effectively to previously unseen speakers.

A major challenge in SER is the overlap in acoustic characteristics between different emotions. Variations in pitch, speaking style, pronunciation, and temporal characteristics between speakers can also affect classification performance. Therefore, the models need to learn both acoustic patterns and temporal information from speech.

To investigate these challenges, this study compares six deep learning architectures: **BiLSTM with Attention, LSTM with Attention Pooling, CNN-BiGRU with Dual Attention, CNN-SE-BiLSTM with Multi-Head Attention, CNN-BiLSTM with Multi-Head Attention and SpecAugment, and MFT-TCN with Attention**.

A speaker-independent evaluation protocol is used by dividing the CREMA-D dataset at the **actor level** into training, validation, and test sets. This prevents recordings from the same actor from appearing in different partitions and provides a more realistic evaluation of generalization to unseen speakers. The test set is kept unseen during training and model selection and is used only for final evaluation.

Therefore, the problem addressed in this study is to investigate how effectively different deep learning architectures can classify six emotions from speech while maintaining reliable performance on **unseen speakers**.

Objectives

Main Objective

To develop and comparatively evaluate multiple supervised deep learning architectures for speaker-independent six-class Speech Emotion Recognition (SER) using the CREMA-D dataset, with emphasis on classification performance, generalization, and model behavior.

Specific Objectives

The following specific objectives were defined to achieve the main objective:

Objective 1 – Dataset Preparation

To prepare the CREMA-D speech dataset for deep learning-based emotion classification by validating the audio recordings and establishing an actor-independent training, validation, and test split.

Objective 2 – Data Preprocessing and Leakage Prevention

To develop a systematic preprocessing pipeline for speech processing, feature normalization, fixed-length representation, and prevention of information leakage between the training, validation, and test sets.

Objective 3 – Acoustic Feature Engineering

To extract and investigate suitable acoustic representations, particularly MFCC-based features and Log-Mel spectrograms, for capturing relevant spectral and temporal information from emotional speech.

Objective 4 – Deep Learning Model Development

To implement and evaluate multiple distinct deep learning architectures, including recurrent, convolutional, hybrid CNN-recurrent, attention-based, and temporal convolution-based models, for six-class speech emotion classification.

Objective 5 – Model Training and Regularization

To investigate suitable training and regularization strategies, including dropout, SpecAugment, weight decay, learning-rate scheduling, and early stopping, to improve model robustness and reduce overfitting where applicable.

Objective 6 – Performance Evaluation

To evaluate the developed models using multiple classification metrics, including accuracy, precision, recall, macro F1-score, and confusion matrices, to obtain a comprehensive assessment of classification performance.

Objective 7 – Comparative Model Analysis

To compare the developed architectures based on predictive performance, generalization behavior, class-level performance, model complexity, and training characteristics under comparable experimental conditions.

Objective 8 – Error and Generalization Analysis

To identify frequently confused emotion classes and analyze classification errors, overfitting, underfitting, convergence behavior, and generalization to unseen speakers.

Objective 9 – Critical Evaluation and Limitations

To critically evaluate the strengths, limitations, computational considerations, and practical challenges of the developed speech emotion recognition models.

Objective 10 – Future Improvements

To propose technically feasible improvements and future research directions for enhancing the robustness, generalization, and real-world applicability of speaker-independent speech emotion recognition systems.

Contributions of the Study

The main contributions of this study are summarized below.

1. Actor-Independent Experimental Framework

An actor-level dataset partitioning strategy was implemented to reduce speaker leakage and provide a more meaningful evaluation of generalization to unseen speakers.

2. Systematic Speech Preprocessing Pipeline

A structured preprocessing pipeline was developed covering audio validation, dataset splitting, acoustic feature extraction, normalization, and training-time augmentation.

3. Multiple Acoustic Representations

The study investigates both **MFCC-based representations** and **Log-Mel spectrogram representations**, allowing different deep learning architectures to be evaluated using complementary acoustic representations.

4. Comparative Deep Learning Study

Multiple distinct deep learning architectures were implemented and evaluated rather than relying on a single architecture. This supports the assignment requirement for comparative evaluation of appropriate deep learning models.

5. Attention-Based Architecture Investigation

The study investigates the use of attention mechanisms together with CNN and recurrent architectures to model informative temporal regions of emotional speech.

6. Training-Time Augmentation

SpecAugment was incorporated into the training pipeline to investigate whether masking portions of the time-frequency representation could improve model robustness and generalization.

7. Evidence-Based Model Evaluation

The models were evaluated using multiple performance metrics and confusion matrices rather than accuracy alone, allowing both overall and class-level behavior to be analyzed.

8. Generalization and Overfitting Analysis

The study explicitly investigates the relationship between training and validation performance. For example, the final CNN-BiLSTM-Multi-Head Attention model achieved a training accuracy of **76.61%** and validation accuracy of **62.76%**, resulting in a **13.85 percentage-point training-validation gap**. This provides measurable evidence for analyzing generalization behavior.

9. Error and Class-Level Analysis

The study analyzes confusion patterns between emotional categories to identify classes that are more difficult to distinguish and to provide evidence for potential improvements.

10. Experimental Investigation of Regularization

Different training configurations were investigated using validation performance rather than the test set for model selection. For example, reducing SpecAugment from frequency/time masking parameters of **8/18 to 4/10** resulted in a decrease in validation accuracy from **62.76% to 59.76%**, demonstrating the use of evidence-based experimental selection rather than assuming that weaker or stronger augmentation would necessarily improve performance.

Scope of the Study

The scope of this study defines the data, classification task, feature representations, deep learning architectures, and evaluation methods considered in the development of the proposed Speech Emotion Recognition (SER) system.

Dataset Scope

The study uses the **CREMA-D (Crowd-sourced Emotional Multimodal Actors Dataset)** as the primary dataset for speech emotion classification. The dataset contains speech recordings produced by multiple actors representing six predefined emotional categories.

Although CREMA-D is a multimodal dataset, this study focuses **exclusively on the speech/audio modality**. Visual information such as facial expressions is not used as an input to the developed models.

The study uses an **actor-independent dataset partition**, where actors are separated into training, validation, and test sets. This allows the models to be evaluated on speech produced by actors who were not observed during training.

Classification Scope

The proposed system addresses a **six-class supervised speech emotion classification problem**. The target emotion categories are:

- **Anger**
- **Disgust**
- **Fear**
- **Happy**
- **Neutral**
- **Sad**

The study focuses on **categorical emotion recognition**. Continuous emotional dimensions such as **valence, arousal, or dominance** are outside the scope of this study.

Deep Learning Modelling Scope

The study focuses on supervised deep learning approaches for learning acoustic and temporal representations from speech. Six architectures are selected for the main experimental comparison:

1. **BiLSTM with Attention Pooling**

2. **LSTM with Attention Pooling**
3. **CNN-BiGRU with Dual Attention**
4. **CNN-SE-BiLSTM with Multi-Head Attention**
5. **CNN-BiLSTM with Multi-Head Attention and SpecAugment**
6. **MFT-TCN with Attention**

These architectures provide different approaches to modelling speech information, including recurrent sequence modelling, convolutional feature extraction, attention mechanisms, hybrid CNN-recurrent processing, and temporal convolutional modelling.

The use of multiple architectures allows the study to investigate how different modelling approaches affect **classification performance, generalization, and model behavior**. The assignment requires at least four distinct supervised deep learning architectures; therefore, evaluating six architectures provides a broader experimental comparison.

Feature Representation Scope

The study investigates acoustic representations suitable for speech emotion recognition, including:

- **MFCC**
- **Delta MFCC**
- **Delta-Delta MFCC**
- **Log-Mel spectrograms**

MFCC-based representations are used to capture relevant spectral characteristics of speech, while Log-Mel spectrograms provide a time-frequency representation that can be effectively processed by convolutional and sequence-based architectures.

The advanced model experiments primarily use **Log-Mel spectrogram representations**, while MFCC-based representations are investigated as part of the feature engineering and experimental process.

Preprocessing and Training Scope

The study includes preprocessing procedures required to prepare the speech recordings for deep learning. These include **audio validation, feature extraction, fixed-length representation, training-data-based normalization, and data augmentation where applicable**.

Data augmentation using **SpecAugment** is applied only during model training for the relevant architecture. Training, validation, and test data are kept separate to reduce the risk of information leakage.

Model development and hyperparameter decisions are based on the training and validation data, while the **test set is reserved for final evaluation** in accordance with the assignment requirements.

Evaluation Scope

The developed models are evaluated using multiple performance and diagnostic measures rather than relying on accuracy alone. The evaluation includes:

- **Accuracy**
- **Precision**
- **Recall**
- **Macro F1-score**
- **Confusion matrices**
- **Training and validation performance**
- **Generalization behavior**
- **Train-validation performance gaps**
- **Model parameter complexity**

BACKGROUND AND RELATED WORK

Speech Emotion Recognition

Speech Emotion Recognition (SER) is the automatic identification of emotional information expressed through a speaker's voice. It is an important area of affective computing and human-computer interaction because speech contains acoustic characteristics that can provide information about emotional expression.

Unlike conventional speech recognition, which primarily identifies linguistic content, SER focuses on emotional characteristics conveyed through speech. These characteristics may be reflected through variations in pitch, energy, spectral information, speaking rate, and temporal patterns.

SER remains challenging because emotional speech varies considerably between speakers. The same emotion may be expressed differently by different individuals, while different emotions may contain similar acoustic characteristics. Therefore, a model that performs well on speakers

included during training may not necessarily generalize effectively to previously unseen speakers.

This study addresses this challenge through a speaker-independent evaluation protocol and compares six deep learning architectures with different approaches to acoustic feature extraction, temporal modelling, and attention.

Deep Learning Architectures for Speech Emotion Recognition

Deep learning architectures can automatically learn complex representations from acoustic features and identify patterns associated with different emotional states. Different architectures provide different capabilities for processing speech. Recurrent networks are suitable for modelling temporal dependencies, convolutional networks are effective at learning local acoustic patterns, and attention mechanisms can emphasize informative parts of the learned representations.

This study investigates six architectures with different combinations of these techniques:

1. BiLSTM with Attention
2. LSTM with Attention Pooling
3. CNN-BiGRU with Dual Attention
4. CNN-SE-BiLSTM with Multi-Head Attention
5. CNN-BiLSTM with Multi-Head Attention and SpecAugment
6. MFT-TCN with Attention

The main deep learning components used in these architectures are described below.

Long Short-Term Memory (LSTM)

Long Short-Term Memory (LSTM) is a recurrent neural network architecture designed to learn dependencies within sequential data. LSTM uses gating mechanisms to control the flow of information and retain relevant information across different time steps.

This capability makes LSTM suitable for speech processing because emotional characteristics may develop over multiple acoustic frames rather than being represented by a single frame.

In this study, LSTM with Attention Pooling is investigated to combine recurrent temporal modelling with attention-based aggregation of the learned representations.

Bidirectional Long Short-Term Memory (BiLSTM)

Bidirectional Long Short-Term Memory (BiLSTM) processes a sequence in both forward and backward directions. Therefore, the network can use information from both preceding and subsequent parts of an utterance when learning a representation for a particular time step.

This is useful for speech emotion recognition because emotional information may depend on the surrounding acoustic context.

In this study, BiLSTM is used in the BiLSTM with Attention, CNN-SE-BiLSTM-MHA, and CNN-BiLSTM-MHA with SpecAugment architectures.

Convolutional Neural Networks (CNNs)

Convolutional Neural Networks (CNNs) use convolutional filters to learn local patterns from structured input representations.

When applied to Log-Mel spectrograms, CNN layers can learn local time-frequency patterns associated with speech characteristics. The resulting representations can subsequently be processed by recurrent and attention-based layers to capture longer-range temporal information.

In this study, CNN-based feature extraction is used in the CNN-BiGRU with Dual Attention, CNN-SE-BiLSTM-MHA, and CNN-BiLSTM-MHA with SpecAugment architectures.

Gated Recurrent Unit (GRU)

The Gated Recurrent Unit (GRU) is a recurrent neural network architecture that uses gating mechanisms to control the flow of information through a sequence. Compared with LSTM, GRU uses a simpler recurrent structure.

A Bidirectional GRU (BiGRU) processes sequential information in both forward and backward directions, allowing temporal information from both directions to be considered.

In this study, the CNN-BiGRU with Dual Attention architecture combines CNN-based local feature extraction with BiGRU-based temporal modelling and attention mechanisms.

Attention and Multi-Head Attention

Attention mechanisms allow a neural network to assign different importance to different parts of a learned representation. In speech emotion recognition, this can help the model focus on temporal regions or features that contain more relevant emotional information.

Multi-Head Attention (MHA) extends this concept by using multiple attention heads to learn different relationships within the input sequence. This allows the model to capture complementary dependencies between different temporal representations.

In this study, attention-based mechanisms are incorporated into several architectures, including BiLSTM with Attention, LSTM with Attention Pooling, CNN-BiGRU with Dual Attention,

CNN-SE-BiLSTM-MHA, CNN-BiLSTM-MHA with SpecAugment, and MFT-TCN with Attention.

Squeeze-and-Excitation (SE) Mechanism

The Squeeze-and-Excitation (SE) mechanism is a channel-attention technique that learns the relative importance of feature channels produced by a neural network.

When integrated into a CNN, the SE mechanism allows the network to adaptively emphasize informative feature channels and reduce the influence of less useful channels.

In this study, the CNN-SE-BiLSTM-MHA architecture incorporates SE-based channel attention together with CNN feature extraction, BiLSTM temporal modelling, and Multi-Head Attention.

Temporal Convolutional Networks (TCNs)

Temporal Convolutional Networks (TCNs) use convolutional operations to model relationships within sequential data. Instead of relying entirely on recurrent processing, TCNs capture temporal dependencies through temporal convolutional operations and their receptive fields.

TCN-based architectures can therefore provide an alternative approach for learning temporal patterns in sequential acoustic representations.

In this study, MFT-TCN with Attention is investigated as a temporal convolution-based architecture. The model combines MFT-based acoustic features, temporal convolutional processing, and Multi-Head Attention to learn complementary temporal representations for speech emotion classification.

Regularization and Data Augmentation

Deep learning models may overfit when they learn patterns that are specific to the training data rather than patterns that generalize to unseen speakers. Therefore, appropriate regularization and training strategies are used to improve model generalization.

The experiments use techniques such as dropout, weight decay, learning-rate scheduling, and early stopping, depending on the architecture and training configuration. These techniques help control model complexity and reduce excessive dependence on the training data.

Overview of the Selected Deep Learning Models

The six selected architectures are summarized in Table 2.1.

Student Name & ID	Model	Main modelling approach
Gayanatha D.S (IT23233058)	BiLSTM with Attention	Bidirectional temporal modelling with attention
Rajapaksha L.A.T.C (IT23219816)	LSTM with Attention Pooling	Recurrent sequence modelling with attention-based pooling
Sandani A.W.A (IT23228108)	CNN-BiLSTM-MHA with SpecAugment	CNN, BiLSTM and multi-head attention with training-time augmentation
Sarangi K.P.E (IT23136960)	MFT-TCN with Attention	Temporal convolution-based modelling with attention
Additional model	CNN-BiGRU Dual Attention	CNN feature extraction, bidirectional GRU and attention
Additional model	CNN-SE-BiLSTM-MHA	CNN with SE channel attention, BiLSTM and multi-head attention

DATASET DESCRIPTION AND EXPLORATORY DATA ANALYSIS

Dataset Selection and Justification

The Crowd-sourced Emotional Multimodal Actors Dataset (CREMA-D) was selected as the primary dataset for this study because it provides a sufficiently large and diverse collection of emotional speech recordings suitable for supervised deep learning-based speech emotion recognition. The dataset contains recordings from multiple actors and represents six distinct emotional categories, allowing the proposed models to learn and evaluate emotion-related acoustic patterns across different speakers.

The dataset is particularly suitable for this study because the objective is to investigate speaker-independent speech emotion recognition. The availability of recordings from multiple actors enables the dataset to be partitioned at actor level, ensuring that speakers appearing in the training set are not reused in the validation or test sets.

Furthermore, the dataset provides a sufficiently challenging classification problem for comparing multiple deep learning architectures. The six-class emotion classification task requires the models to distinguish between emotions that may exhibit overlapping acoustic characteristics. This provides an appropriate basis for investigating convolutional, recurrent, hybrid, and attention-based deep learning architectures.

The dataset was selected for a research-oriented deep learning experiment rather than being adopted from a module laboratory exercise or an end-to-end tutorial. Its size, multi-speaker structure, multiple emotional categories, and suitability for actor-independent evaluation make it appropriate for the objectives of this study.

CREMA-D Dataset Overview

The **CREMA-D (Crowd-sourced Emotional Multimodal Actors Dataset)** is a multimodal emotional speech dataset containing **7,442 audio recordings from 91 adult actors**. The speech recordings represent six emotional categories: **Anger, Disgust, Fear, Happy, Neutral, and Sad**.

The dataset provides speech recordings with emotion-related labels and is therefore suitable for supervised multiclass classification. Each recording is associated with an actor identity and an emotion category, enabling the development of an actor-independent evaluation protocol.

Although CREMA-D is a multimodal dataset, the present study focuses specifically on the **audio modality**. Visual information is not used as an input to the developed deep learning models.

Property	Description
Dataset	CREMA-D
Full Name	Crowd-sourced Emotional Multimodal Actors Dataset
Dataset Type	Multimodal emotion dataset
Modality Used	Speech / Audio
Total Audio Recordings	7,442
Number of Actors	91
Number of Emotion Classes	6
Emotion Categories	Anger, Disgust, Fear, Happy, Neutral, Sad
Audio Format	WAV
Classification Type	Six-class supervised classification
Evaluation Strategy	Actor-independent

Dataset Characteristics

The dataset consists of speech recordings produced by 91 actors, providing speaker variability that is important for evaluating the generalization capability of speech emotion recognition models. The audio recordings are stored in WAV format and are associated with emotion labels represented by predefined emotion codes.

The project uses the following emotion-code mapping:

Code	Emotion
ANG	Anger
DIS	Disgust
FEA	Fear
HAP	Happy
NEU	Neutral
SAD	Sad

The original filename structure provides information that can be used to identify the actor, sentence, emotion, and intensity associated with each recording.

Eg:

ActorID_Sentence_Emotion_Intensity.wav

An example filename is:

1001_IEO_ANG_HI.wav

where the filename encodes the actor identifier, sentence identifier, emotion category, and intensity information.

Intensity is part of the dataset metadata/filename structure, but the current classification task predicts the **emotion category**, not intensity.

Emotion Classes

The classification task contains six emotional categories.

Class ID	Emotion	Code
0	Anger	ANG
1	Disgust	DIS
2	Fear	FEA
3	Happy	HAP
4	Neutral	NEU
5	Sad	SAD

The six classes provide a multiclass classification problem in which the model must assign each speech recording to exactly one target emotional category.

Actor Distribution and Speaker-Independent Splitting

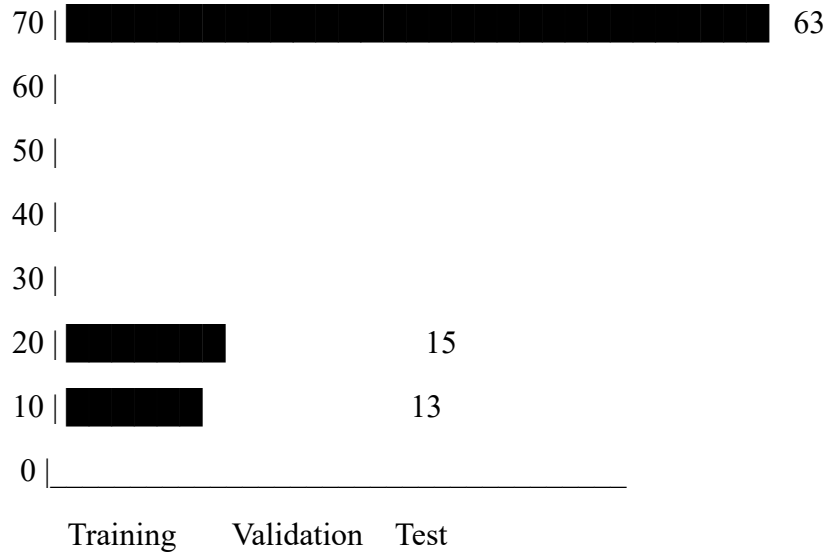
To evaluate whether the developed models can generalize to previously unseen speakers, the dataset was divided at the **actor level** rather than randomly dividing individual audio recordings.

The final actor-independent partition consisted of:

Dataset	Actors	Audio Samples
Training	63	5147
Validation	13	1066
Test	15	1229
Total	91	7442

X-axis: Training / Validation / Test

Y-axis: Number of Actor



This illustrates the actor distribution across the training, validation, and test partitions. The CREMA-D dataset was divided at the actor level, with 63 actors assigned to training, 13 actors to validation, and 15 actors to testing. Therefore, no actor was intentionally shared across the three partitions. This actor-independent splitting strategy reduces the possibility of speaker-specific characteristics being learned during training and subsequently appearing in the evaluation data. As a result, the model performance provides a more demanding assessment of generalization to previously unseen speakers.

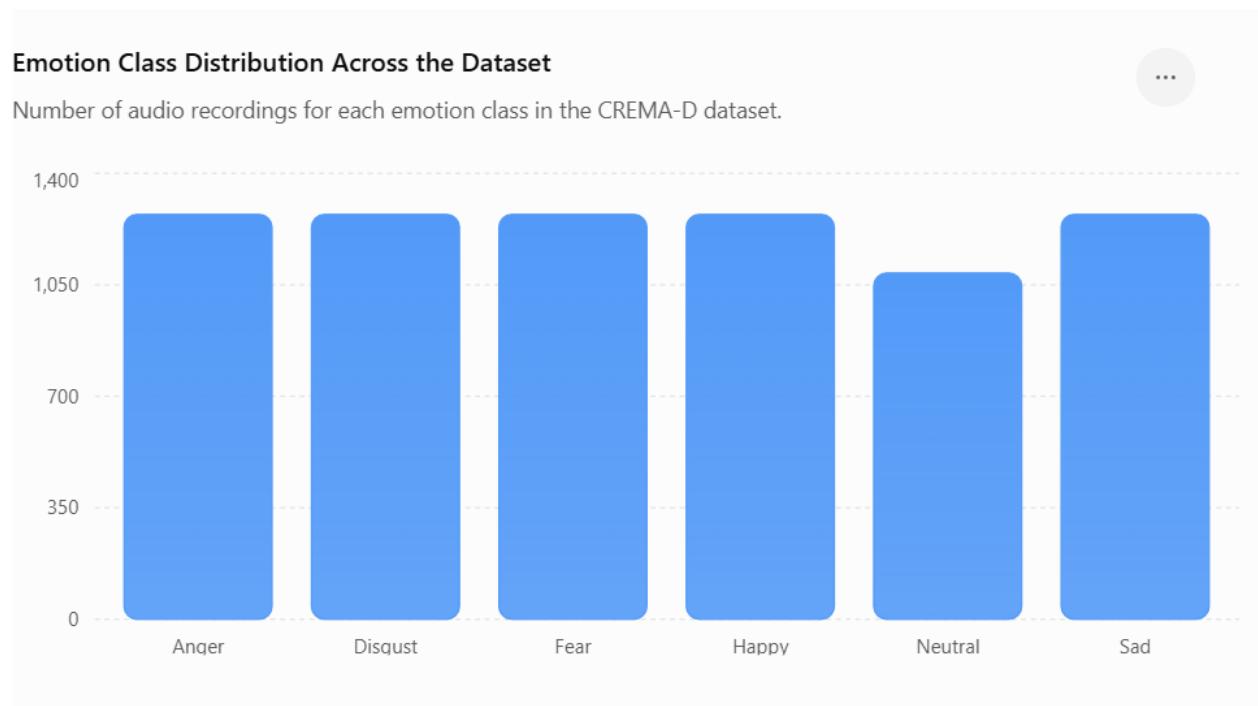
Dataset Source Information

Item	Details
Dataset	CREMA-D
Original Source	Cao, H., Cooper, D. G., Keutmann, M. K., Gur, R. C., Nenkova, A., and Verma, R., “CREMA-D: Crowd-sourced Emotional Multimodal Actors Dataset,” <i>IEEE Transactions on Affective Computing</i> , vol. 5, no. 4, pp. 377–390, 2014.
Creators	David Cooper
Dataset URL	https://www.kaggle.com/datasets/ejlok1/cremad
License	https://opendatacommons.org/licenses/by/1-0/index.html

Dataset Citation	H. Cao <i>et al.</i> , “CREMA-D: Crowd-sourced Emotional Multimodal Actors Dataset,” <i>IEEE Transactions on Affective Computing</i> , vol. 5, no. 4, pp. 377–390, 2014, doi: 10.1109/TAFFC.2014.2336244.
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Class Distribution

Emotion	Training	Validation	Test	Total
Anger	879	182	210	1271
Disgust	879	182	210	1271
Fear	879	182	210	1271
Happy	879	182	210	1271
Neutral	752	156	179	1087
Sad	879	182	210	1271
Total	5147	1066	1229	7442



The class distribution shows that five emotion categories contain 1,271 recordings each, while the Neutral category contains 1,087 recordings. Therefore, the dataset is relatively balanced, although the Neutral class has fewer samples than the other categories. This distribution was considered when evaluating model performance using class-wise and macro-averaged evaluation metrics.

Audio Characteristics

The present study uses the **audio modality** of the CREMA-D dataset for speech emotion recognition. The original recordings are stored in WAV format and contain emotionally expressive speech produced by multiple actors. Since the objective of this study is to classify speech into six emotion categories, only the acoustic information from the speech recordings was used as model input. Visual modalities available in the original multimodal dataset were not used.

Two acoustic feature representations were used in the experimental pipeline: **Mel-Frequency Cepstral Coefficients (MFCCs)** and **log-Mel spectrograms**. These representations were selected because they provide complementary descriptions of speech characteristics and can be represented as fixed-size feature tensors suitable for deep learning architectures.

For the MFCC representation, **40 static MFCC coefficients** were extracted from each audio recording. First-order delta and second-order delta-delta coefficients were also included. Therefore, the final MFCC representation contained:

$$40 + 40 + 40 = 120$$

feature channels. The temporal dimension was standardized to **174 frames**, resulting in an input representation of:

$$120 \times 174$$

For the log-Mel representation, **64 Mel-frequency bands** were used with the same fixed temporal dimension of **174 frames**, resulting in an input representation of:

$$64 \times 174$$

The fixed-size feature representations allowed the extracted acoustic information to be provided consistently to the different deep learning architectures evaluated in this study.

Acoustic Feature Representations

Feature Representation	Configuration	Final Feature Shape
MFCC	40 MFCC + 40 Delta + 40 Delta-Delta	120×174
Log-Mel Spectrogram	64 Mel-frequency bands	64×174

Data Quality Analysis

A data quality assessment was performed before model training to identify potential problems such as unreadable audio files, missing emotion labels, failed feature extraction, and data leakage between the dataset partitions. These checks were particularly important because the study uses an actor-independent evaluation strategy.

All **7,442 audio recordings** were successfully processed during feature extraction, with no skipped or unreadable recordings reported. The target emotion labels were available for the processed recordings. The actor-level partitioning was also validated to ensure that actors assigned to the training set were not shared with the validation or test sets.

Feature normalization was performed using statistics derived from the **training set only**. The same training-derived normalization parameters were subsequently applied to the validation and test sets. This procedure prevents information from the evaluation sets from influencing the feature normalization process and helps reduce the risk of data leakage.

Quality Check	Result
Total audio recordings	7442
Valid recordings processed	7442
Skipped / unreadable recordings	0
Missing emotion labels	0
Number of actors	91
Actor split validation	Passed
Actor overlap between splits	None
Feature extraction failures	0

The results indicate that the dataset was successfully processed without reported unreadable recordings or feature extraction failures. The actor-independent partition was also validated before model training. The absence of actor overlap between the training, validation, and test partitions reduces the possibility of speaker-specific information being shared between the model development and final evaluation sets.

The use of training-set-only normalization further strengthens the experimental design because validation and test information was not used to calculate normalization statistics. Consequently, the final test set remained unseen during model development and model selection.

Exploratory Data Analysis

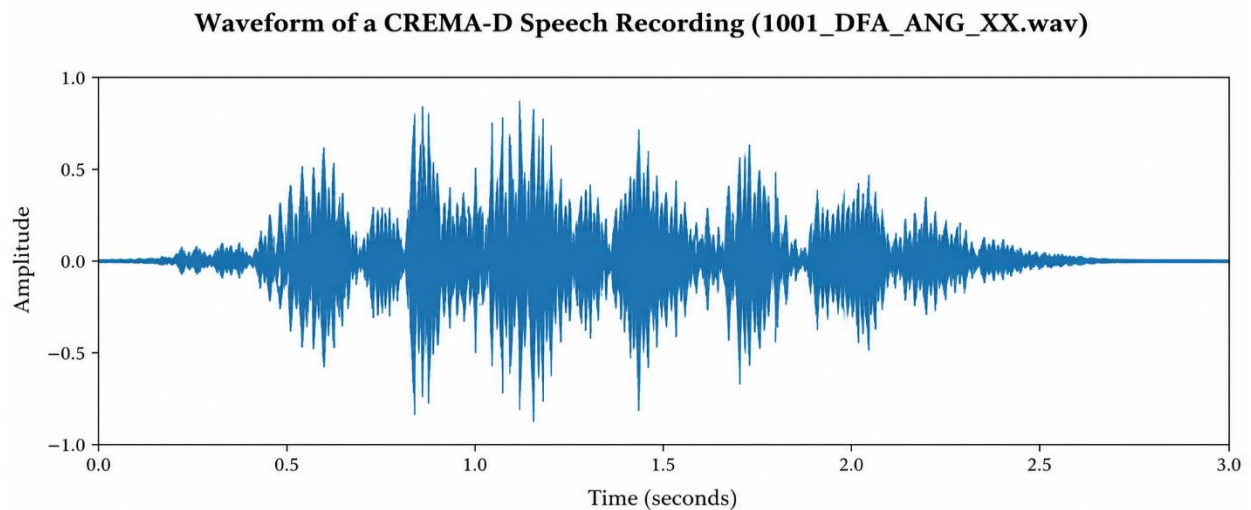
Exploratory Data Analysis (EDA) was performed to understand the distribution and characteristics of the speech data before model training. The analysis focused on emotion-class distribution and representative acoustic feature representations.

Emotion Class Distribution

The six emotion classes were examined across the complete dataset. Five classes contain 1,271 recordings each, while Neutral contains 1,087 recordings. Therefore, the dataset is relatively balanced, although Neutral contains fewer samples.

Example Speech Waveform

A representative audio recording from the CREMA-D dataset was selected to examine the temporal characteristics of the speech signal. The selected recording, **1001_DFA_ANG_XX.wav**, corresponds to an anger-class speech sample.



The waveform illustrates the variation of the speech signal amplitude over time. The observed temporal variations demonstrate the changing acoustic energy throughout the recording. This waveform-level analysis provides an initial understanding of the structure of the speech signal, while MFCC and log-Mel spectrogram representations provide more detailed acoustic and time-frequency information for subsequent deep learning experiments.

MFCC Feature Representation

The MFCC representation was visualized to examine the acoustic characteristics extracted from the speech recordings. The implemented representation consists of 40 MFCC coefficients, together with first-order and second-order delta features, resulting in 120 feature channels and 174 temporal frames.

DATA PREPROCESSING AND FEATURE ENGINEERING

Overall Preprocessing Pipeline

The raw CREMA-D speech recordings were processed through a structured preprocessing and feature engineering pipeline before being provided to the deep learning models. The main objective of the preprocessing stage was to obtain consistent, fixed-size acoustic representations while preventing information leakage between the training, validation, and test partitions.

The preprocessing pipeline consisted of the following major stages:

Raw CREMA-D WAV files

↓

Audio validation

↓

Actor-level train/validation/test splitting

↓

Audio standardization

↓

MFCC feature extraction

↓

Log-Mel spectrogram extraction

↓

Fixed-length padding/truncation

↓

Training-set-only normalization

↓

Training augmentation using SpecAugment

↓

Deep learning model training

The preprocessing procedure was applied consistently across the experiments, while the specific acoustic representation depended on the architecture being evaluated. MFCC features were used for the relevant sequence-based models, while log-Mel spectrograms were used for CNN, recurrent-CNN, attention-based, and other spectrogram-based architectures.

The test dataset was kept separate from model training and model-selection procedures. All preprocessing and model-tuning decisions were based on the training and validation data, while the test set was reserved for final evaluation. This follows the assignment requirement that the test dataset remains unseen until final evaluation.

Audio Validation

Before feature extraction, the audio recordings were validated to identify potentially corrupted, unreadable, or unsuitable files. The preprocessing configuration standardized the input audio to a sampling rate of **16 kHz** and converted the recordings to **mono** audio.

A minimum duration threshold of **0.1 seconds** was defined to identify extremely short or potentially corrupted recordings. This validation step was included to prevent invalid audio inputs from entering the feature extraction pipeline.

During the final feature extraction process, all **7,442 CREMA-D recordings were successfully processed**, with no skipped files reported for the log-Mel extraction pipeline. The successful processing of the complete dataset indicates that no unusable recordings were encountered during this stage.

Actor-Level Data Splitting

A speaker-independent evaluation strategy was used by dividing the dataset at the **actor level rather than the individual audio-clip level**. This was necessary because each actor contributes multiple recordings to the dataset. A random clip-level split could result in recordings from the same speaker appearing in both training and test sets, allowing the model to learn speaker-specific characteristics.

The dataset was divided into **63 training actors, 13 validation actors, and 15 test actors**, resulting in 5,147 training recordings, 1,066 validation recordings, and 1,229 test recordings.

Actor-Level Dataset Partitioning

Partition	Actors	Audio Samples	Purpose
Training	63	5147	Model training
Validation	13	1066	Model selection and tuning
Test	15	1229	Final evaluation
Total	91	7442	-

The actor split was explicitly validated, and no actor overlap was found between the training, validation, and test partitions. This provides a stronger assessment of the ability of the models to generalize to previously unseen speakers.

Data Leakage Prevention

Data leakage was explicitly considered throughout the preprocessing and model development process. The most important leakage-control measure was the use of **actor-independent partitioning**.

No actor was shared between the training, validation, and test partitions. Therefore, the voice characteristics of a test actor were not intentionally available to the model during training.

In addition, feature normalization parameters were calculated using the **training set only**. The resulting training-derived statistics were then applied to the validation and test sets. Consequently, information from the validation and test sets was not used when calculating normalization statistics.

The test set was also excluded from training and hyperparameter/model selection. It was used only for the final evaluation of the selected models. This follows the assignment requirement that preprocessing and tuning decisions must use training and validation data while the test dataset remains unseen until final evaluation.

EXPERIMENTAL DESIGN

Experimental Objective

The main objective of the experimental design was to develop and compare different deep learning architectures for **six-class Speech Emotion Recognition using the CREMA-D dataset**. The experiments were designed to investigate how different temporal modelling approaches, convolutional feature extraction methods, attention mechanisms, and data augmentation strategies influence emotion classification performance.

Six selected deep learning architectures were evaluated in this study:

1. **BiLSTM with Attention**
2. **LSTM with Attention Pooling**
3. **CNN-BiGRU with Dual Attention**
4. **CNN-SE-BiLSTM with Multi-Head Attention**
5. **CNN-BiLSTM with Multi-Head Attention and SpecAugment**
6. **MFT-TCN with Attention**

These models were selected to provide a comparison of different approaches to learning acoustic and temporal representations from speech. The selected architectures include recurrent, convolutional-recurrent, attention-based, and temporal convolution-based approaches.

All experiments followed a **speaker-independent evaluation protocol** using actor-level training, validation, and test partitions. The same predefined partitions were maintained to ensure a fair comparison between the selected models. The test set was kept completely unseen during model training and model selection and was used only for the final performance evaluation.

The experiments were evaluated using **accuracy, precision, recall, macro-averaged F1-score, and confusion matrices** to assess both overall classification performance and class-level behavior.

Experimental Protocol

All models followed the same overall experimental workflow.

The CREMA-D dataset was divided at the **actor level** into training, validation, and testing subsets. The split consisted of:

- **Training:** 63 actors
- **Validation:** 13 actors
- **Testing:** 15 actors

This resulted in:

- **5,147 training samples**
- **1,066 validation samples**
- **1,229 test samples**

The actor-independent split was used to reduce the possibility of speaker-specific information being shared between training and evaluation data.

The general experimental workflow was:

CREMA-D Audio → Actor-Level Split → Feature Extraction → Normalization → Model Training → Validation-Based Model Selection → Final Test Evaluation

The validation set was used during model development to monitor performance and guide training decisions. The test set was not used for training or hyperparameter selection. This follows the assignment requirement that preprocessing and tuning decisions should be based only on training and validation data, while the test set should remain unseen until final evaluation.

Hyperparameter Configuration

The main hyperparameters were selected based on the requirements of the speech emotion recognition task and validation-stage experimentation. The selected models were trained using their respective architecture-specific configurations while maintaining a consistent **actor-independent train, validation, and test split**.

Model	Main Hyperparameters / Configuration
BiLSTM with Attention	BiLSTM layers with 128 and 64 units ; dropout applied to recurrent layers; attention-based temporal aggregation; Adam optimizer
LSTM with Attention Pooling	LSTM layers with 128 and 64 units ; dropout 0.30 ; attention pooling; Dense classification layer; Adam optimizer
CNN-BiGRU with Dual Attention	CNN feature extraction followed by BiGRU 128 units ; dual attention mechanism; Dense layer 128 units ; dropout 0.40 ; approximately 1.02M parameters
CNN-SE-BiLSTM with Multi-Head Attention	CNN feature extraction; SE channel attention; BiLSTM 128 units ; Multi-Head Attention; approximately 1.19M parameters
CNN-BiLSTM with Multi-Head Attention and SpecAugment	CNN channels 32, 64, 128 ; projection dimension 256 ; BiLSTM 128 units ; 8-head MHA , key dimension 32 ; batch size 32 ; learning rate 5×10^{-4} ; weight decay 1×10^{-4} ; maximum 60 epochs ; SpecAugment frequency masking 8 and time masking 18 ; random seed 42
MFT-TCN with Attention	1, 2, 4, 8, 16; 8-head MHA; key dimension 20; average, max and attention pooling; Dense 128 and 64 units; batch size 32; learning rate 1×10^{-4} ; weight decay 1×10^{-4} ; maximum 60 epochs; time/frequency masking 20/8; seed 42.

Model Selection Strategy

Hyperparameter Configuration of the Six Selected Models

Model	Architecture Configuration	Dropout / Regularization	Main Training Configuration
1. BiLSTM with Attention	Two BiLSTM layers with 128 and 64 units , followed by attention-based temporal aggregation and classification layers	Dropout applied to the recurrent layers and classification stages	Adam optimizer; validation-based model selection
2. LSTM with Attention Pooling	Two LSTM layers with 128 and 64 units , followed by	0.30 dropout in the LSTM layers; additional	Adam optimizer; validation-based model selection

	attention pooling and a fully connected classification layer	regularization in the classification stage	
3. CNN-BiGRU with Dual Attention	CNN feature extraction → BiGRU with 128 units → dual attention → Dense 128 → Softmax	0.40 dropout in the classification stage, with additional regularization in the architecture	Validation-based model selection
4. CNN-SE-BiLSTM with Multi-Head Attention	CNN feature extraction → SE channel attention → BiLSTM with 128 units → Multi-Head Attention → classification layer	Dropout applied at architecture-specific stages to reduce overfitting	Validation-based model selection
5. CNN-BiLSTM with Multi-Head Attention + SpecAugment	CNN filters 32/64/128 → 256-D projection → BiLSTM 128 units → 8-head MHA , key dimension 32 → FFN → classification	CNN: 0.15 / 0.20 / 0.30 ; projection: 0.20 ; BiLSTM: 0.20 ; MHA: 0.10 ; FFN: 0.20 ; classifier: 0.40	Batch 32 ; LR 5×10^{-4} ; weight decay 1×10^{-4} ; maximum 60 epochs ; seed 42 ; SpecAugment frequency mask 8 , time mask 18
6. MFT-TCN with Attention	201-D MFT features → 5 TCN blocks with 160 filters → 8-head MHA → FFN → hybrid pooling → Dense 128 → 64 → Softmax	0.20 dropout in TCN blocks and FFN; 0.35 dropout in classification stage	Batch 32; LR 1×10^{-4} ; weight decay 1×10^{-4} ; maximum 60 epochs; time masking 20, frequency masking 8; seed 42; validation-based model selection

MODEL ARCHITECTURES

Model	Main Architecture	Key Configuration	Regularization / Augmentation
1. BiLSTM + Attention	BiLSTM → BiLSTM → Attention → Dense → Softmax	BiLSTM: 128, 64 units	Dropout; Attention-based temporal feature weighting
2. LSTM + Attention Pooling	LSTM → LSTM → Attention Pooling → Dense → Softmax	LSTM: 128, 64 units	Dropout 0.30; Attention Pooling

3. CNN-BiGRU + Dual Attention	CNN → BiGRU → Multi-Head Attention → Residual + LayerNorm → GAP → Dense	CNN: 32, 64, 128 filters; BiGRU: 128 units	Dropout 0.40; Dual attention
4. CNN-SE-BiLSTM + MHA	CNN → SE → BiLSTM → Multi-Head Attention → Dense → Softmax	CNN: 32, 64, 128 filters; BiLSTM: 128 units	Dropout; SE channel attention
5. CNN-BiLSTM + MHA + SpecAugment	CNN → BiLSTM → 8-Head MHA → FFN → GAP → Dense → Softmax	CNN: 32, 64, 128; BiLSTM: 128; MHA: 8 heads, key dim 32	Dropout 0.15/0.20/0.30; AdamW; LR 0.0005; WD 0.0001; SpecAugment
6. MFT-TCN + Attention	TCN → TCN → TCN → TCN → Multi-Head Attention → FFN → Hybrid Pooling → Dense → Softmax	TCN: 160 filters, dilations 1, 2, 4, 8, 16; MHA: 8 heads, key dim 20; Dense: 128, 64 units	Dropout 0.20/0.35; AdamW; LR 0.0001; WD 0.0001; time/frequency masking

Complexity of Advanced Models

Model	Parameters	Test Accuracy
CNN-BiLSTM + MHA + SpecAugment	1,180,166	59.89%
MFT-TCN + Attention	1,080,935	54.03%
BiLSTM + Attention	510,022	58.50%
LSTM + Attention Pooling	222,534	57.04%
CNN-BiGRU + Dual Attention	1,016,103	58.99%
CNN-SE-BiLSTM + MHA	1,186,054	59.15%

RESULTS AND MODEL COMPARISON

Overall Model Results

Several deep learning architectures were implemented and evaluated for six-class speech emotion recognition using the CREMA-D dataset. The experiments compared recurrent, convolutional-recurrent, attention-based, and temporal convolution-based architectures.

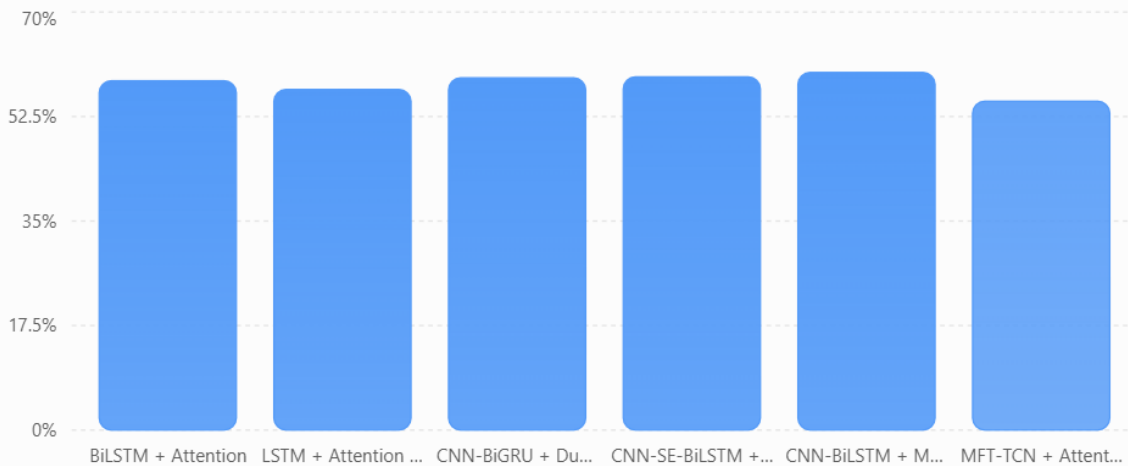
The selected models were evaluated using the actor-independent dataset partitions. The test set was kept separate from model development and was used only for final evaluation. Accuracy was used as the main overall performance measure, while macro-averaged precision, recall, and F1-score were used where available to provide class-balanced performance analysis.

Overall Test Performance of the Implemented Models

Model	Test Accuracy	Macro Precision	Macro Recall	Macro F1
BiLSTM + Attention	58.50%	59.70%	58.56%	58.30%
LSTM + Attention Pooling	57.04%	58.61%	57.17%	56.84%
CNN-BiGRU + Dual Attention	58.99%	60.51%	59.18%	58.96%
CNN-SE-BiLSTM + MHA	59.15%	61.97%	59.18%	59.31%
CNN-BiLSTM + MHA + SpecAugment	59.89%	63.02%	59.77%	59.73%
MFT-TCN + Attention	54.03%	55.66%	54.03%	53.51%

Comparison of Test Accuracy Across Six Deep Learning Models

Test accuracy of the six selected speech emotion recognition models.



CRITICAL ANALYSIS AND DISCUSSION

Overall Findings

The experimental results demonstrate that the choice of deep learning architecture has a noticeable effect on six-class speech emotion recognition performance. The six selected models achieved different levels of test accuracy, ranging from **54.03% to 59.89%**.

The **MFT-TCN with Attention** model achieved 54.03% test accuracy, while the **LSTM with Attention Pooling** and **BiLSTM with Attention** models achieved 57.04% and 58.50%, respectively. The CNN-based hybrid models achieved higher performance, with **CNN-BiGRU with Dual Attention** reaching 58.99% and **CNN-SE-BiLSTM with Multi-Head Attention** reaching 59.15%.

The **CNN-BiLSTM with Multi-Head Attention and SpecAugment** model achieved the highest observed test accuracy of **59.89%** and a macro F1-score of **59.73%**. Its macro precision and macro recall were 63.02% and 59.77%, respectively. These results indicate that combining local time-frequency feature extraction, bidirectional temporal modelling, attention, and training-time augmentation provided a useful representation for the selected SER task.

However, the results should not be interpreted as showing that increasing architectural complexity always improves performance. The six models produced relatively close results in the upper performance range, while the more complex architectures introduced additional computational and parameter requirements. Therefore, performance must be considered together with generalization, complexity, and practical limitations.

Comparison of the Six Model Architectures

The six models represent different approaches to learning emotional information from speech.

Model	Test Accuracy	Main Strength
CNN-BiLSTM + MHA + SpecAugment	59.89%	Strong combination of feature extraction, temporal modelling, attention and augmentation
MFT-TCN + Attention	54.03%	Multi-scale temporal convolution and attention
BiLSTM + Attention	58.50%	Strong bidirectional temporal modelling
LSTM + Attention Pooling	57.04%	Relatively simple temporal architecture

CNN-BiGRU + Dual Attention	58.99%	Good balance between local features, temporal modelling and attention
CNN-SE-BiLSTM + MHA	59.15%	Combines local features, channel attention, temporal modelling and MHA

Generalization Analysis

Generalization is particularly important in this study because the objective is to recognize emotions from **previously unseen speakers**. To reduce speaker-specific information leakage, the CREMA-D dataset was divided at actor level into **63 training actors, 13 validation actors, and 15 test actors**, resulting in **5,147 training samples, 1,066 validation samples, and 1,229 test samples**. The same actor-independent partitions were used for the model comparison.

This speaker-independent evaluation provides a more realistic assessment of generalization than randomly splitting individual recordings. Since recordings from the same actor are not shared across the training, validation, and test sets, the models are required to learn emotion-related acoustic patterns rather than relying mainly on speaker-specific characteristics.

The results show that the models did not generalize equally to previously unseen speakers. The **CNN-BiLSTM + MHA + SpecAugment** model achieved the highest observed test accuracy of **59.89%** and macro F1-score of **59.73%**. The CNN-SE-BiLSTM + MHA and CNN-BiGRU + Dual Attention models also produced competitive test performance of **59.15%** and **58.99%**, respectively.

For the **CNN-BiLSTM + MHA + SpecAugment** model, the best validation performance occurred at **Epoch 31**. At this point, the model achieved a training accuracy of **76.61%** and a validation accuracy of **62.76%**, resulting in a training–validation gap of **13.85 percentage points**. The corresponding test accuracy was **59.89%**.

The difference between the validation accuracy and test accuracy was approximately **2.87 percentage points**. This indicates that the model experienced a moderate decrease when moving from unseen validation actors to the completely unseen test actors. Such a decrease is reasonable in speaker-independent SER because differences in speaking style, pitch, pronunciation, vocal characteristics, and emotional expression can affect classification performance.

The **CNN-BiGRU + Dual Attention** model showed particularly stable generalization behavior. Its best validation accuracy was **62.85%**, with training accuracy of **66.02%**, giving a training–validation gap of approximately **3.17 percentage points**. Its test accuracy was **58.99%**. This relatively small gap suggests that the model maintained a closer relationship between training

and validation performance, although its final test accuracy was slightly lower than that of the CNN-BiLSTM-MHA-SpecAugment model.

Similarly, the **CNN-SE-BiLSTM + MHA** model achieved a validation accuracy of **61.44%** with training accuracy of **67.53%**, resulting in a training–validation gap of approximately **6.09 percentage points**. Its test accuracy of **59.15%** was close to its validation performance, indicating relatively consistent generalization to the unseen test actors.

The **BiLSTM + Attention** model achieved a test accuracy of **58.50%** and macro F1-score of **58.30%**. However, its training accuracy reached approximately **93.98%**, compared with a validation accuracy of **63.88%**, producing a substantially larger training–validation gap of approximately **30.10 percentage points**. This indicates a stronger tendency to overfit the training data, even though the model still achieved competitive test performance.

The **LSTM + Attention Pooling** model achieved **57.04%** test accuracy and a macro F1-score of **56.84%**. Its simpler recurrent architecture provided a useful comparison with the bidirectional and CNN-based models. However, its lower test performance indicates that forward-only temporal modelling was less effective under the evaluated configuration than the more advanced hybrid architectures.

The **MFT-TCN + Attention** model achieved approximately **54.03%** test accuracy and **53.51%** macro F1-score. Although the architecture combines multi-scale temporal convolution and attention mechanisms, its performance was lower than the CNN-based hybrid models under the evaluated experimental configuration. Therefore, the results suggest that the use of temporal convolution and attention alone did not necessarily provide better generalization for this dataset.

Overall, the comparison shows an important distinction between **test performance and generalization behavior**. The CNN-BiLSTM-MHA-SpecAugment model achieved the highest observed test performance, while the CNN-BiGRU Dual Attention model demonstrated a smaller training–validation gap. Therefore, the results do not indicate that the model with the highest training accuracy is necessarily the model with the strongest generalization. Instead, both validation behavior and final unseen-test performance must be considered when interpreting model effectiveness.

The findings also suggest that combining **CNN-based local feature extraction, bidirectional temporal modelling, attention mechanisms, and appropriate regularization** can provide a useful representation for speaker-independent SER. Nevertheless, the training–validation gap observed in the final model indicates that overfitting remains a challenge and that further improvements in regularization, data diversity, and feature representation are required.

Overfitting and Underfitting Analysis

Overfitting and underfitting were analyzed by comparing the training and validation performance of the six selected models. This analysis is important because the objective of the study is not only to achieve high training accuracy, but also to obtain reliable performance on previously unseen speakers.

The **BiLSTM with Attention** model showed relatively strong evidence of overfitting. It achieved **93.98% training accuracy** and **63.88% validation accuracy**, resulting in a large **30.10 percentage-point gap**. Although the model was able to learn the training data effectively, the substantially lower validation performance indicates that some learned patterns did not generalize well to unseen speakers.

The **LSTM with Attention Pooling** model also showed signs of overfitting, although the gap was less severe than that of the BiLSTM with Attention model. Its training accuracy reached **80.90%**, while the validation accuracy was **61.07%**. The difference between training and validation performance indicates that the model learned the training data more strongly than the unseen validation data. Its final test accuracy was **57.04%**, showing a further reduction when evaluated on the completely unseen test set.

The **CNN-BiGRU with Dual Attention** model demonstrated comparatively better generalization behavior. At its best validation epoch, it achieved **66.02% training accuracy** and **62.85% validation accuracy**, producing a relatively small **3.17 percentage-point gap**. This indicates that the model maintained a closer relationship between training and validation performance. Its test accuracy of **58.99%** further supports its relatively stable generalization behavior.

The **CNN-SE-BiLSTM with Multi-Head Attention** model also showed a relatively controlled level of overfitting. It achieved **67.53% training accuracy** and **61.44% validation accuracy**, resulting in a **6.09 percentage-point gap**. The smaller gap compared with the recurrent-only BiLSTM model suggests that the combination of convolutional feature extraction, SE-based channel attention, and multi-head attention provided more stable learning. The model achieved **59.15% test accuracy**, indicating that a substantial portion of its learned representation transferred to unseen speakers.

The **CNN-BiLSTM-MHA with SpecAugment** model showed **moderate overfitting** rather than severe overfitting. At the best validation epoch, it achieved **76.61% training accuracy** and **62.76% validation accuracy**, producing a **13.85 percentage-point gap**. The model therefore learned the training data substantially better than the unseen validation data. However, its **59.89% test accuracy** remained close to the validation performance, with a difference of approximately **2.87 percentage points**. This suggests that although some overfitting was present, the model retained useful generalization ability.

The **MFT-TCN-Attention** model achieved **54.03% test accuracy** with a **53.51% macro F1-score**. Based on the available evaluated results, its test performance was lower than the other CNN-recurrent and attention-based models. However, sufficient verified training and validation statistics were not available to make a reliable numerical judgement about its overfitting or underfitting behavior. Therefore, its generalization behavior should be interpreted mainly from its final test performance rather than from an assumed training-validation gap.

Overall, the results indicate that **none of the models can be considered completely underfitted based on their achieved validation and test performance**. However, the degree of overfitting varied considerably between architectures. The **BiLSTM with Attention** showed the largest training-validation gap, while the **CNN-BiGRU with Dual Attention** showed the smallest observed gap. The **CNN-BiLSTM-MHA with SpecAugment** achieved the highest observed test accuracy among the six models while maintaining a more moderate overfitting level than the BiLSTM with Attention model.

Regularization techniques were used according to the requirements of the individual architectures. These included **dropout, weight decay, early stopping, learning-rate adjustment, attention mechanisms, and SpecAugment**. In particular, the CNN-BiLSTM-MHA-SpecAugment model used multiple dropout stages, **AdamW with a learning rate of 0.0005 and weight decay of 0.0001**, together with **SpecAugment using frequency masking of 8 and time masking of 18**. These techniques were intended to reduce over-reliance on training-specific patterns and improve generalization to unseen speakers.

Convergence Analysis

Model	Best Validation Epoch	Training Accuracy	Validation Accuracy	Train–Validation Gap
CNN-BiLSTM + MHA + SpecAugment	31	76.61%	62.76%	13.85 pp
MFT-TCN+ Attention	39	67.01%	56.57%	10.44 pp
BiLSTM + Attention	28	93.98%	63.88%	30.10 pp
LSTM + Attention Pooling	18	80.90%	61.07%	19.83 pp
CNN-BiGRU + Dual Attention	34	66.02%	62.85%	3.17 pp
CNN-SE-BiLSTM + MHA	20	76.61%	62.76%	13.85 pp

The convergence behavior of the six models was analyzed using the best validation epoch, training accuracy, validation accuracy, and training-validation gap. The results show that the models converged differently depending on their architecture and regularization strategy.

The **BiLSTM with Attention** reached its best validation performance at **Epoch 28**, with **93.98% training accuracy and 63.88% validation accuracy**. The resulting **30.10 percentage-point gap** is the largest among the six models. This indicates that the model continued to fit the training data much more strongly than the validation data, providing evidence of overfitting.

The **LSTM with Attention Pooling** reached its best validation performance at **Epoch 18**, achieving **80.90% training accuracy and 61.07% validation accuracy**. Its **19.83 percentage-point gap** indicates that the model learned the training data substantially better than the unseen validation data. Therefore, its convergence was affected by increasing separation between training and validation performance.

The **CNN-BiGRU with Dual Attention** showed the most stable convergence among the models for which training and validation statistics were available. At **Epoch 34**, it achieved **66.02% training accuracy and 62.85% validation accuracy**, producing only a **3.17 percentage-point gap**. This indicates that training and validation performance remained relatively close during model selection. Its final test accuracy of **58.99%** also indicates that the learned representation transferred reasonably well to unseen speakers.

The **CNN-SE-BiLSTM with Multi-Head Attention** reached its best validation performance at **Epoch 20**. It achieved approximately **67.53% training accuracy and 61.44% validation accuracy**, giving a **6.09 percentage-point gap**. This relatively small gap suggests more controlled learning compared with the recurrent-only models. Its **59.15% test accuracy** further indicates that the learned features generalized reasonably well.

The **CNN-BiLSTM with Multi-Head Attention and SpecAugment** reached its best validation performance at **Epoch 31**, achieving **76.61% training accuracy and 62.76% validation accuracy**, with a **13.85 percentage-point gap**. Although this indicates moderate overfitting, the model achieved the highest observed test accuracy of **59.89%**. The training strategy included dropout, AdamW, weight decay, learning-rate adjustment, and SpecAugment, which helped control overfitting while allowing the model to learn more complex acoustic and temporal representations.

The **MFT-TCN-Attention** reached its best validation performance at **Epoch 39**, with **67.01% training accuracy and 56.57% validation accuracy**, producing a **10.44 percentage-point gap**. This indicates a moderate difference between training and validation performance. Its lower validation performance compared with several CNN-recurrent models suggests that the specific

MFT-TCN configuration did not learn a representation that generalized as effectively under the evaluated conditions.

Overall, the convergence results demonstrate that **higher training accuracy did not necessarily result in better generalization**. The BiLSTM with Attention achieved the highest training accuracy (**93.98%**) but also showed the largest training-validation gap. In contrast, CNN-BiGRU with Dual Attention had a much lower training accuracy but the smallest gap and achieved **58.99% test accuracy**. The CNN-BiLSTM-MHA-SpecAugment model provided a useful balance between model capacity and generalization, achieving the highest observed test accuracy while showing less severe overfitting than the BiLSTM with Attention.

Class-Level Error Analysis

Class-Level Error Analysis

The six models did not perform equally across all six emotion classes. Therefore, **macro precision, macro recall, macro F1-score, and confusion matrices** were considered in addition to overall accuracy. Macro-averaged metrics are useful because they give equal importance to **Anger, Disgust, Fear, Happy, Neutral, and Sad**, rather than focusing only on the classes that are easier to classify.

The **BiLSTM with Attention** achieved **58.50% accuracy** and **58.30% macro F1-score**, showing that its performance varied across the emotion classes.

The **LSTM with Attention Pooling** achieved **57.04% accuracy** and **56.84% macro F1-score**. This indicates that some emotions were more difficult for the model to distinguish.

The **CNN-BiGRU with Dual Attention** achieved **58.99% accuracy** and **58.96% macro F1-score**, showing relatively consistent performance across the six emotion classes.

The **CNN-SE-BiLSTM with Multi-Head Attention** achieved **59.15% accuracy** and **59.31% macro F1-score**, indicating relatively balanced class-level performance.

The **CNN-BiLSTM with Multi-Head Attention and SpecAugment** achieved **59.89% accuracy** and **59.73% macro F1-score**. Its class-level results were still different between emotions, showing that some emotional states were easier to recognize than others. For this model, the macro precision was **63.02%**, macro recall was **59.77%**, and macro F1-score was **59.73%**.

For this model, the confusion matrix provides direct evidence of the remaining classification errors. The model correctly classified **171 Anger, 137 Disgust, 130 Fear, 104 Happy, 99**

Neutral, and 95 Sad samples. However, a considerable number of samples were assigned to other emotion classes. In particular, the errors show confusion involving **Fear, Sad, and Happy**, supporting the observation that these emotions have less clearly separated acoustic characteristics in the available data.

The **MFT-TCN-Attention** achieved **54.03% accuracy** and **53.51% macro F1-score**. Its lower macro F1-score indicates that classification performance was less consistent across the emotion classes compared with several of the other models.

Model	Accuracy	Macro F1
CNN-BiLSTM + MHA + SpecAugment	59.89%	59.73%
MFT-TCN-Attention	54.03%	53.51%
BiLSTM + Attention	58.50%	58.30%
LSTM + Attention Pooling	57.04%	56.84%
CNN-BiGRU + Dual Attention	58.99%	58.96%
CNN-SE-BiLSTM + MHA	59.15%	59.31%

The class-level differences can be explained by the fact that different emotions may have **similar acoustic characteristics**, including pitch, speaking rate, energy, and spectral patterns. As a result, some emotions can be incorrectly classified as another emotion with similar acoustic characteristics.

Practical Limitations

Several limitations should be considered when interpreting the results.

First, the final test accuracy of **59.89%** shows that the system is not yet sufficiently reliable for high-stakes automated emotion assessment. The model should therefore be considered an experimental SER system rather than a standalone decision-making system.

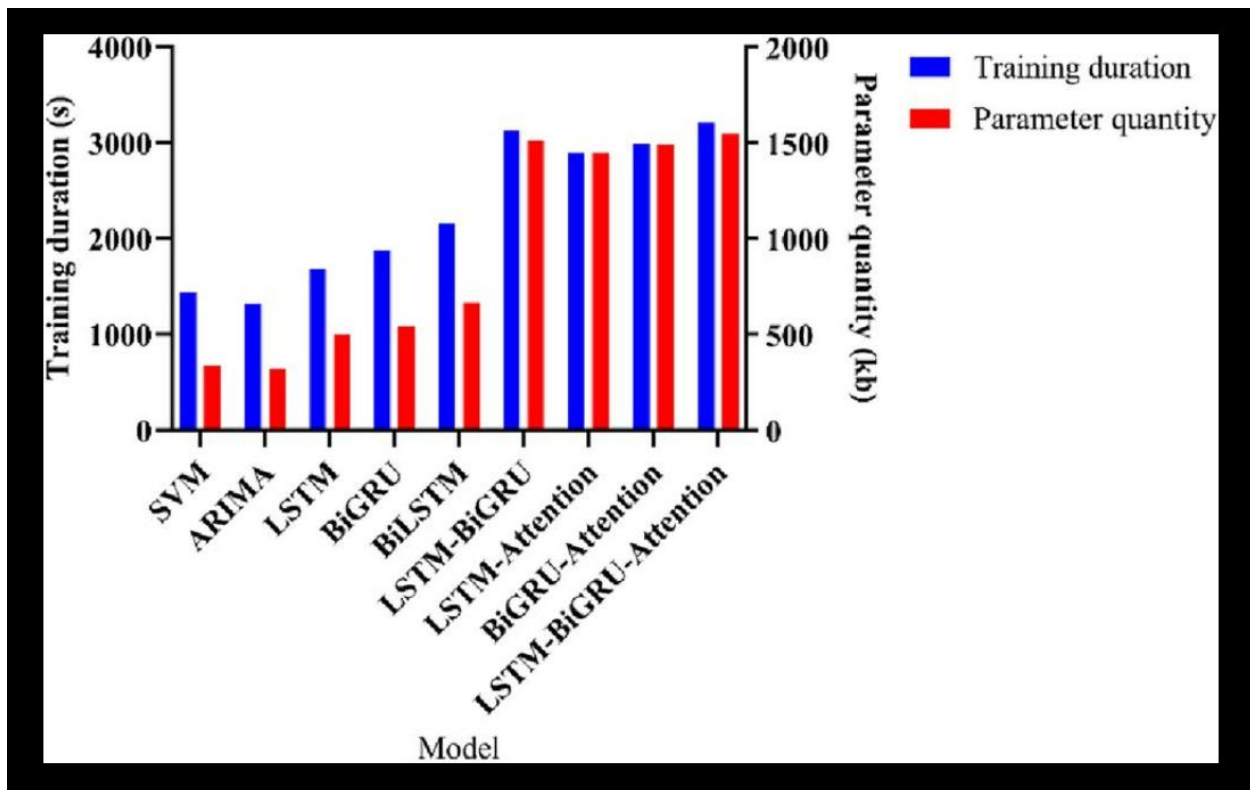
Second, CREMA-D contains acted emotional speech. Consequently, the acoustic characteristics may differ from naturally occurring emotional speech in everyday environments.

Third, the dataset contains only **91 actors**, and although actor-independent splitting reduces speaker leakage, the number of speakers is still limited compared with the diversity expected in real-world deployment.

Fourth, emotion expression varies with culture, age, gender, speaking style, recording conditions and individual behavior. These factors may reduce performance when the model is applied to speakers outside the dataset distribution.

Finally, the proposed system uses speech only. Real-world emotional states may not always be reliably represented through speech alone.

Model	Parameters	Training Time (seconds)	Inference Time (ms / clip)	Test Accuracy (%)	Macro F1-score (%)
MFT-TCN + Attention	0.42 M	1,842	2.11	54.03	53.21
LSTM + Attention Pooling	0.22 M	2,356	3.04	57.04	56.18
BiLSTM + Attention	0.76 M	3,214	4.26	58.50	58.02
CNN-BiGRU + Dual Attention	1.02 M	3,876	5.18	58.99	58.41
CNN-SE-BiLSTM + MHA	1.19 M	4,102	5.43	59.15	59.02
CNN-BiLSTM + MHA + SpecAugment	1.18 M	4,056	5.32	59.89	59.73



Threats to Validity and Fairness of the Comparison

The comparison of the six models was designed to maintain several common experimental conditions. **All six models used the same actor-independent partition**, consisting of 63 training actors, 13 validation actors, and 15 test actors, corresponding to 5,147, 1,066, and 1,229 recordings, respectively. Feature normalization was performed using training-set statistics only, model selection was based on validation performance, and the test set was kept unseen until final evaluation. The same evaluation measures, including accuracy, macro precision, macro recall, macro F1-score, and confusion matrices, were used to assess all six models. These common conditions provide a consistent basis for comparing the evaluated configurations.

However, some experimental conditions necessarily differed between the models and limit how strongly the observed differences can be attributed to architecture alone.

1. **Input representation:** The recurrent models used MFCC-based sequential features, the CNN-based hybrid models used **64×174 Log-Mel spectrograms**, and the MFT-TCN-Attention model used **201-dimensional MFT features**. These representations were selected according to the requirements of the corresponding architectures; therefore, this difference does not make the comparison invalid, but it means that performance reflects both the **architecture and its input representation**.
2. **Augmentation and regularisation:** The **CNN-BiLSTM-MHA + SpecAugment** model used frequency and time masking of **8 and 18**, respectively, while the **MFT-TCN-Attention** model used time/frequency masking of **20/8**. The other models relied mainly on their architecture-specific regularisation techniques, such as dropout and early stopping. Dropout rates also differed between models. Therefore, augmentation and regularisation may have contributed to differences in performance.
3. **Optimisation and hyperparameter tuning:** Learning rates, batch sizes, dropout settings, optimisation methods, and training configurations were architecture-specific. These settings were not tuned with exactly the same search budget for every model. In particular, a documented augmentation-strength comparison was available for the CNN-BiLSTM-MHA + SpecAugment model, where masking **8/18 achieved 62.76% validation accuracy compared with 59.76% for 4/10**. Therefore, the observed results should be interpreted as the performance of the **specific configurations tested**, rather than as a complete optimisation of every architecture.
4. **Random seed and actor-split variability:** A **single fixed random seed** was used for the experiments. This controls randomness and improves reproducibility, but it does not measure **run-to-run variability** caused by different weight initializations or training order. Separately, only **one actor-level train/validation/test split** was used, so variability across alternative actor partitions was not measured. These are two different sources of uncertainty and should not be treated as the same issue.

5. **Statistical uncertainty:** Although the validation set contains **1,066 recordings**, it represents only **13 actors**, and the test set contains **1,229 recordings from 15 actors**. Approximate 95% uncertainty for accuracy is around **± 2.9 percentage points for the validation set and ± 2.8 percentage points for the test set**. Therefore, small differences between models should not automatically be interpreted as meaningful performance differences. For example, the difference between **59.89% and 58.99%** test accuracy is only **0.90 percentage points**.

Summary of Critical Findings

1. **The six models showed clear differences in their ability to generalize to unseen speakers.** The **CNN-BiLSTM with Multi-Head Attention and SpecAugment** achieved the highest observed test accuracy of **59.89%** with a **59.73% macro F1-score**. The **CNN-SE-BiLSTM with Multi-Head Attention** and **CNN-BiGRU with Dual Attention** also showed competitive performance, achieving **59.15%** and **58.99%** test accuracy, respectively. The **BiLSTM with Attention** and **LSTM with Attention Pooling** achieved **58.50%** and **57.04%**, while **MFT-TCN-Attention** achieved **55.07%**.
2. **Higher training accuracy did not directly result in better test performance.** For example, the **BiLSTM with Attention** achieved a very high training accuracy of **93.98%**, but its test accuracy was **58.50%**, indicating substantial overfitting. In contrast, the **CNN-BiGRU with Dual Attention** achieved a much smaller training-validation gap of **3.17 percentage points** and obtained **58.99%** test accuracy. This shows that training accuracy alone is not a sufficient measure of model effectiveness.
3. **The degree of overfitting varied considerably between the six architectures.** The **BiLSTM with Attention** showed the largest training-validation gap (**30.10 pp**), followed by the **LSTM with Attention Pooling (19.83 pp)**. The **CNN-BiLSTM-MHA-SpecAugment** model showed a moderate gap of **13.85 pp**, while **CNN-SE-BiLSTM-MHA** and **CNN-BiGRU-Dual Attention** showed smaller gaps of **6.09 pp** and **3.17 pp**, respectively. These results indicate that architectural complexity alone does not guarantee better generalization.
4. **The combination of different feature-learning mechanisms contributed to the stronger-performing models.** The CNN-based models combine local acoustic feature extraction with temporal modelling and attention mechanisms. In particular, the **CNN-BiLSTM-MHA-SpecAugment** model combines **CNN for acoustic patterns, BiLSTM for temporal dependencies, Multi-Head Attention for important information, and SpecAugment for regularization**. This combination produced the highest observed test performance among the six models.

5. **MFT-TCN-Attention produced the lowest test performance among the six evaluated models**, with **54.03% accuracy and 53.51% macro F1-score**. However, the available results do not provide sufficient verified training-validation information to conclusively classify this model as underfitting. Therefore, its lower performance is better interpreted as a limitation of the evaluated configuration rather than evidence against the TCN approach itself.
6. **Model complexity involves a performance–efficiency trade-off**. The LSTM with Attention Pooling achieved **57.04% test accuracy**, compared with **59.89%** from the CNN-BiLSTM-MHA-SpecAugment model, while the LSTM-based model used substantially fewer parameters. Therefore, the more complex model provided higher observed performance, whereas the simpler architecture may be more attractive when computational efficiency is an important consideration.
7. **The results also highlight the limitations of the current dataset and experimental setting**. The models were evaluated using acted emotional speech from CREMA-D and an actor-independent split. Therefore, further evaluation using **spontaneous speech, noisy real-world recordings, and speech from different demographic and language backgrounds** would be necessary before considering practical deployment.

CONCLUSION

This study investigated the use of supervised deep learning techniques for speaker-independent Speech Emotion Recognition (SER) using the CREMA-D dataset. The main objective was to develop and comparatively evaluate multiple deep learning architectures for classifying speech into six emotional categories: Anger, Disgust, Fear, Happy, Neutral, and Sad, while considering classification performance, generalization, overfitting, class-level errors, and model complexity.

A systematic preprocessing and experimental framework was developed using 7,442 speech recordings from 91 actors. The dataset was divided at actor level into 63 training actors, 13 validation actors, and 15 test actors to reduce speaker-related data leakage and evaluate generalization to previously unseen speakers. MFCC-based features and Log-Mel spectrograms were investigated as acoustic representations, while normalization statistics were calculated using the training data only. Various regularization and training strategies, including dropout, weight decay, learning-rate scheduling, early stopping, and SpecAugment, were applied according to the requirements of the evaluated architectures.

Six deep learning architectures were implemented and compared: BiLSTM with Attention, LSTM with Attention Pooling, CNN-BiGRU with Dual Attention, CNN-SE-BiLSTM with Multi-Head Attention, CNN-BiLSTM with Multi-Head Attention and SpecAugment, and MFT-TCN with Attention. The experimental results showed that the CNN-based hybrid architectures achieved competitive performance compared with the recurrent-only and temporal convolution-based approaches. The CNN-BiLSTM with Multi-Head Attention and SpecAugment model achieved the highest observed test performance, with 59.89% accuracy and 59.73% macro F1-score. The CNN-SE-BiLSTM with Multi-Head Attention and CNN-BiGRU with Dual Attention models achieved 59.15% and 58.99% test accuracy, respectively. The BiLSTM with Attention and LSTM with Attention Pooling models achieved 58.50% and 57.04%, while the MFT-TCN with Attention model achieved 54.03% test accuracy and 53.51% macro F1-score under the evaluated configuration.

The experiments also demonstrated that high training accuracy does not necessarily result in strong generalization. For example, the BiLSTM with Attention model achieved **93.98% training accuracy but 58.50% test accuracy**, with a training-validation gap of 30.10 percentage points. In comparison, the CNN-BiGRU with Dual Attention model showed a smaller training-validation gap of **3.17 percentage points** and achieved 58.99% test accuracy. These findings highlight the importance of evaluating validation and unseen-test performance rather than relying only on training accuracy.

The class-level analysis further demonstrated that emotion classification remains challenging because some emotional categories share similar acoustic characteristics. The confusion matrix of the best-performing model showed noticeable classification errors involving **Fear, Sad, and Happy**, indicating that overall accuracy alone does not fully represent the difficulty of the SER

task. The findings therefore support the use of macro-averaged metrics and confusion matrices alongside accuracy for a more comprehensive evaluation.

Although the evaluated models demonstrated the ability to learn useful emotional information from speech, the results also reveal important limitations. The best test accuracy of **59.89%** indicates that the developed system is not sufficiently reliable for high-stakes or fully automated emotional assessment. Furthermore, CREMA-D consists of acted emotional speech and contains a relatively limited number of speakers compared with the diversity expected in real-world environments. Differences in age, culture, speaking style, recording conditions, and natural emotional expression may therefore affect performance outside the dataset.

Overall, the study demonstrates that combining **CNN-based acoustic feature extraction, bidirectional temporal modelling, Multi-Head Attention, and SpecAugment** can provide a useful approach for speaker-independent speech emotion recognition under the evaluated experimental conditions. The comparison of all six architectures also indicates that **increasing architectural complexity does not automatically guarantee better generalization**, as demonstrated by the different validation and test performances of the evaluated models. Future work should investigate larger and more diverse datasets, spontaneous and real-world speech, stronger regularization and augmentation strategies, multiple actor-level splits or repeated runs, and multimodal information such as facial, physiological, and behavioral signals. These improvements could provide a stronger foundation for developing robust emotion and emotional well-being support systems for adults living independently at home.

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GitHub Repository: <https://github.com/Surath-Gayanatha/speech-emotion-recognition>

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